SMART INDIA HACKATHON 2026

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- **Problem Statement ID – SIH26034**
- **Problem Statement Title - Automated Legal Metrology Compliance System**
- **Theme-Miscellaneous**
- **PS Category- Software**
- **Team ID-**
- **Team Name - SITARA**

AI-BASED COMPLIANCE CHECKER FOR PACKAGED COMMODITIES

Sitara

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WHAT OUR SYSTEM DOES

- **AI-based label analysis**
  Detects relevant label regions (e.g., MRP, net quantity, manufacturer) using OpenCV and YOLO.
- **Intelligent text extraction**
  Uses OCR with BERT/NER to extract and classify label text.
- **Compliance validation**
  Checks declarations as per LMPC Rules, 2011.
- **Reports and dashboard**
  Generates PDF reports and maintains inspection history.

HOW IT ADDRESSES THE PROBLEM

- **Reduces manual effort**
  Automates scanning and extraction of label information.
- **Detects missing or incorrect declarations**
  Flags issues in manufacturer details, net quantity, MRP, dates and consumer-care information.
- **Validates compliance**
  Verifies details against LMPC Rules, 2011 and amendments.
- **Ensures visual compliance**
  Checks font size, readability and declaration placement.

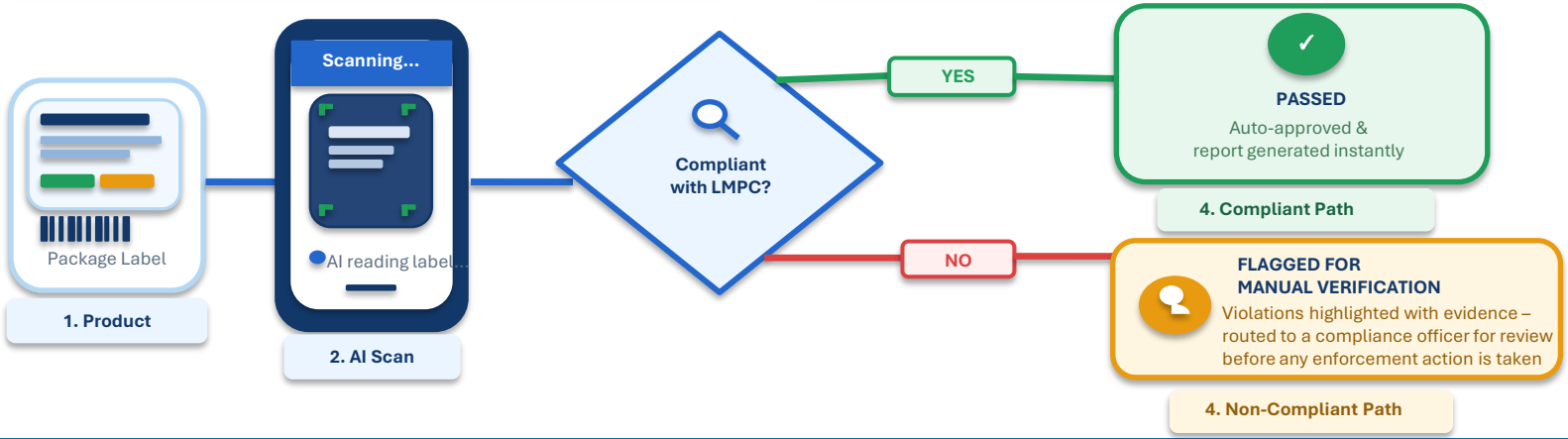
INNOVATION AND UNIQUENESS

- **End-to-end automated inspection**
  Image → Verdict, reducing manual inspection.
- **Region-aware OCR**
  YOLO localizes relevant zones for higher accuracy.
- **Rule-as-Data architecture**
  Editable LMPC rules database.
- **AI + deterministic validation**
  Combines CV, OCR, NER and rule-based reasoning.
- **Explainable and evidence-linked**
  Violation + Rule + Image evidence.

Prototype

Video

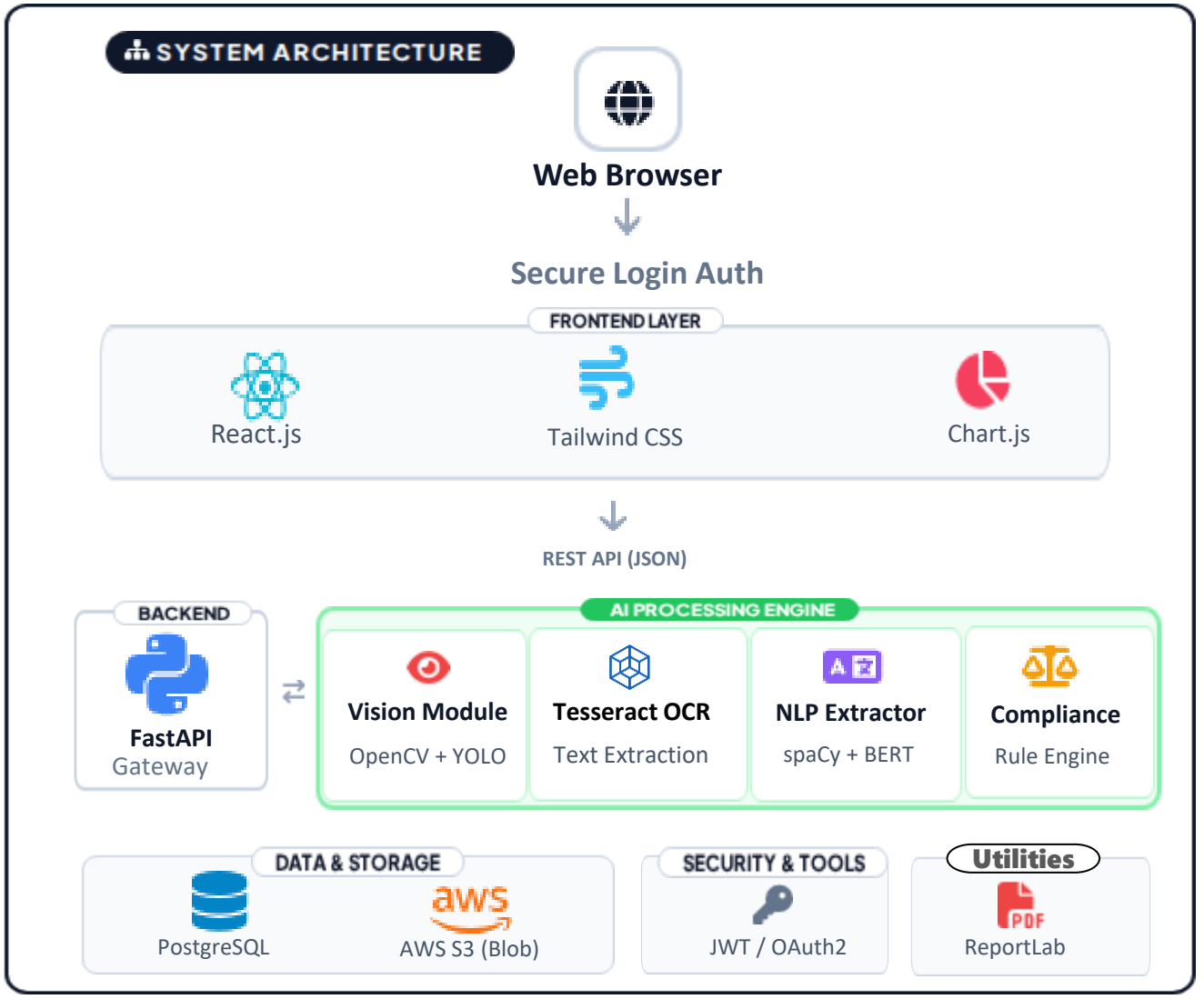
GitHub



TECHNICAL APPROACH

PROJECT METHODOLOGY

1. **Rule digitization:** convert LMPC 2011 clauses into a structured, editable rule database.
2. **Image capture & preprocessing:** user scans package via app; OpenCV corrects skew/blur/lighting.
3. **Label region detection:** YOLO-based model locates declaration zones on the package.
4. **OCR extraction:** Tesseract OCR pulls text from each detected region.
5. **NLP classification:** extracted text tagged into declaration types (MRP, net qty, mfg date, address, care details).
6. **Compliance validation:** rule engine cross-checks flags missing/non-compliant/misleading entries.
7. **Report generation:** auto-generate PDF/editable compliance report with annotated image evidence.
8. **Repository & dashboard:** store scan + report in DB; dashboard shows inspection stats, violation trends, officer-wise activity.



FEASIBILITY AND VIABILITY

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1 CAPTURE
Product Image (Web/Mobile)

2 ANALYSE
Extract & Validate (AI Engine)

3 COMPLY
Result & Report Generation

ANALYSIS OF FEASIBILITY

- **Proven CV stack:** Uses proven computer vision with robust entity extraction mapping.

OpenCV | YOLO | TESSERACT | BERT/NER

- **No special hardware:** Works with standard smartphone cameras and web interfaces.
- **Fully auditable:** Every check traces back to an exact, digitized LMPC clause.

POTENTIAL CHALLENGES

- **OCR accuracy:** Low light, blurred packaging, and complex layouts can hurt extraction.
- **Language variance:** Multilingual text and varied fonts across products.
- **Frequent rule changes:** Legal Metrology updates and commodity-specific requirements.
- **Ambiguous labels:** Incomplete or obscured declarations on packaging.

STRATEGIES TO OVERCOME

- **Smart preprocessing:** Image preprocessing, YOLO region detection, confidence thresholding.
- **Multilingual NLP:** Multilingual OCR with context-aware entity extraction.
- **Version-controlled rules:** Rule engine with effective dates for easy regulatory updates.

IMPACT AND BENEFITS

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REAL-WORLD ENFORCEMENT WORKFLOW

Enforcement officers scan product packages via smartphone app → AI engine automatically extracts & validates LMPC declarations → Generates digital audit reports in seconds.

Scan → AI Check → Instant Audit

OPERATIONAL IMPACT

- **Faster Inspections:** OCR and rule-matching automate label checks.
- **High-Precision Detection:** YOLO and Tesseract reliably flag missing, incorrect declarations.
- **Standardized Enforcement:** One digitized rule base applied uniformly checks.
- **Actionable Analytics:** Dashboards show violation hotspots and compliance trends.

STAKEHOLDER BENEFITS

- **Stronger Consumer Protection:** Automatically verifies MRP, net quantity and manufacturer details.
- **Improved Inspector Productivity:** Less paperwork and data entry, more inspections per shift.
- **Digital Audit Trails:** Report embeds photo evidence, time stamps and compliance status.
- **Scalable & Sustainable:** Cloud repositories replace physical records for nationwide monitoring.

MANUAL VS. OUR SYSTEM

FEATURE	MANUAL	OUR SYSTEM
Automated image scanning	✗	✓
Missing declaration detection	✓	✓
MRP / net quantity check	✓	✓
Digital PDF reports	✗	✓
Centralized inspection log	✗	✓

RESEARCH AND REFERENCES

RESEARCH PAPERS & CASE STUDIES

- **FSSAI Food Label Compliance & Automated Retail Enforcement Study:** evaluation of packaged commodity compliance standards and enforcement outcomes across retail markets.
- **Consumer Awareness & Package Transparency Survey (NCAER / FSSAI):** assessing consumer knowledge regarding mandatory package declarations, MRP, and net quantity disclosures.
- **Tesseract OCR & Ultralytics YOLO Performance Evaluation:** benchmarking accuracy on curved packaging, low-light retail conditions, and multilingual text extraction.

STATUTORY & TECHNICAL REFERENCES

- **The Legal Metrology Act, 2009 & LMPC Rules, 2011:** Ministry of Consumer Affairs, Food and Public Distribution, Government of India. Mandates rules for MRP, manufacturer details, and net quantity.
- **Chhabra et al. (2025):** “Checking Compliance of a Packaged Product”, International Patent Application Publication WO 2025/022453 A1, World Intellectual Property Organization.
- **Open-Source Vision & NLP Frameworks:** OpenCV Foundation (Geometric Preprocessing), Tesseract OCR Engine, and spaCy + BERT for Named Entity Recognition (NER).